UNIT-1 INDUSTRIAL PHARMACY

ONESHOT CRASH COURSE

B PHARMA 5TH SEMESTER

7814870917

IMPORTANT QUESTIONS

M.M.I

1. Discuss in details about concepts and importance of Preformulation studies.
   - Intro, goals and objectives, physio-chemical properties.
   - Application of preformulation consideration in the development of dosage forms (solids, Liquids, parenteral d.f.).

Physical [Solubility (method of enhancing), dissociation constant etc] + Chemical properties

2. Define BCS. How do we classify BCS? What is the significance of BCS classification. M.M.I

Industrial Pharmacy:- It is the branch of pharmacy that deal with the large-scale manufacturing, development, quality control and marketing of pharmaceutical products.

Oneshot Pharma

① Discuss in details about Preformulation studies.

- Intro, goals and objectives, Physio-chemical properties.
- Application of preformulation consideration in the development of dosage forms (solid, liquids, parenteral).

Preformulation studies

These are scientific investigations carried out before developing the final dosage form of a drug.

- They involve studying the physical, chemical and biological properties of the drug substance.

Pre + Formulation

↓ ↓

Before to make

- The aim is to collect essential information that will guide the design, formulation and development of a safe, stable and effective dosage form.

- To know the drug well before making its final form, so that the medicine is safe, effective, stable and economical.

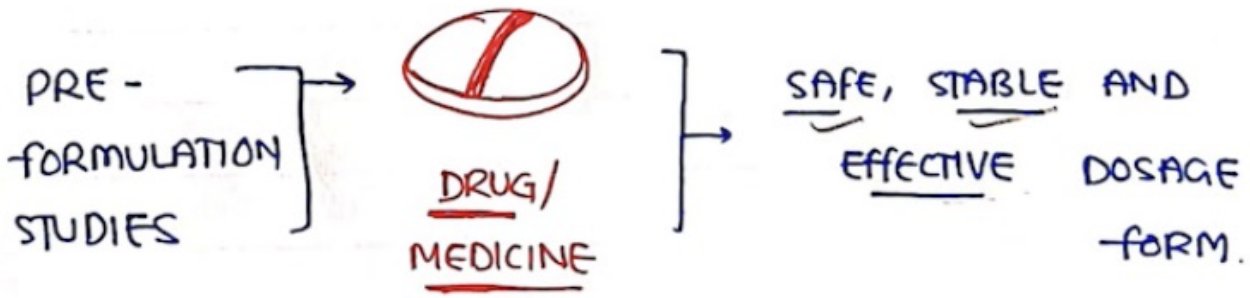
CAREWELL PHARMA

Objective -

- To determine the physiochemical properties of the new drugs.
- To determine compatibility with excipients.
- To establish the drug's solubility and dissolution profile.
- To identify the drug's stability profile.
- To determine its kinetic and stability of drugs.
- To aid in the selection of appropriate dosage form and route of administration.

Goals -

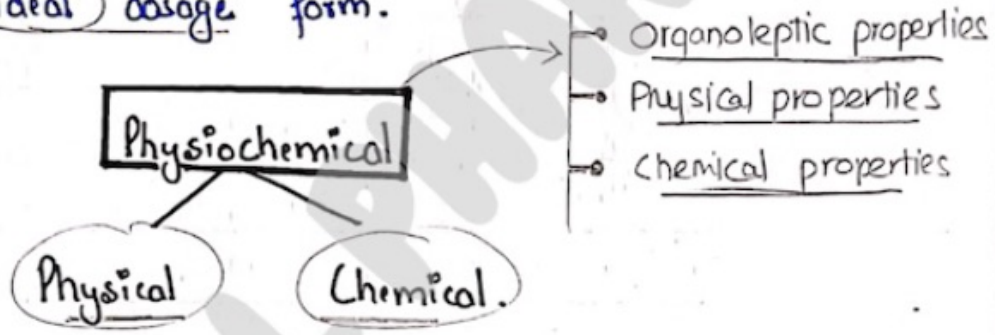
- To develop safe, stable and effective dosage form.
- Maximize drug stability during manufacturing and storage.
- Select suitable excipients for formulation.
- Identify potential problems early to save time and cost.
- Provide data for large-scale manufacturing.
- Improve bioavailability (drug reaching bloodstream in required amount).



Colonial Pharma

Physiochemical characteristics of drug substances

These are the Physical and chemical properties of an API (Active Pharmaceutical Ingredient) that influence its formulation, stability, bioavailability and therapeutic efficacy. These properties are studied during preformulation to design an ideal dosage form.



Physical Properties

⇒ **Organoleptic properties** :- These are those properties of a substance that can perceived by the sense organs. i.e. sight, smell, taste, touch and hearing.

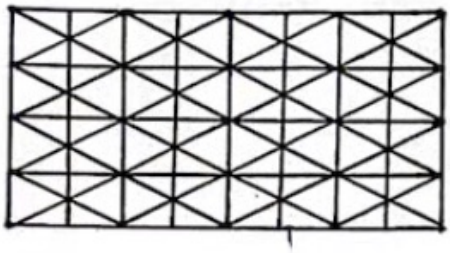
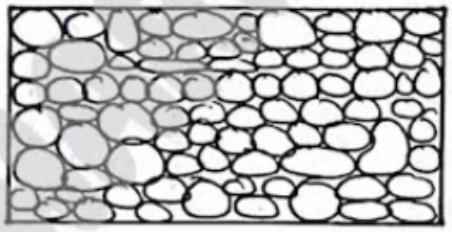
eg. colour, odour, Taste, Texture.

(eg) PCM → white

⇒ **Physical state** - Drug can be solid, liquid or gas.

- Mostly drugs are solid (powder form).
- Solids can be :-
  - Crystalline (ordered structure, stable).
  - Amorphous (no order, more stable but less stable).

7814870917 CHEMICAL PHARMA

Crystalline Form	Amorphous form
• A crystalline drug has molecules arranged in a regular, repeating pattern (like a grid or lattice).	• An Amorphous drug has molecules arranged in a random or disordered way, like a glass.
	
• Easily identified by (X)-ray diffraction (XRPD). • Usually less soluble • Sharp Melting point. [39-40°41] • More stable (chemically & Physically). Example - Paracetamol (stable) (Eg).	• Shows a broad halo in (XRPD) (not sharp peaks). • Usually (more) soluble. • No sharp Melting point (softens over a range). [35°-40°-45°] • Less stable (can degrade faster). Example - Amorphous insulin.

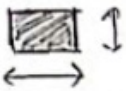
• The crystalline form of (Penicilling) is more stable than Amorphous form.

⇒ **Particle size** :- The particle size of a drug substances means the diameter or dimension of a single particle or group of particles.

• They affects solubility, dissolution, bioavailability.

Eg. Particle (size) ↓ ⟶ Solubility ↑

Easy to mix + affects flow properties



- Small particle (size) (fine powder) is also susceptible to attack by heat, atmosphere, light or humidity which further cause (stability) issues.
- In bulk form (at constant volume) particle size also affects surface area.

Particle size ↓ = Surface Area ↑

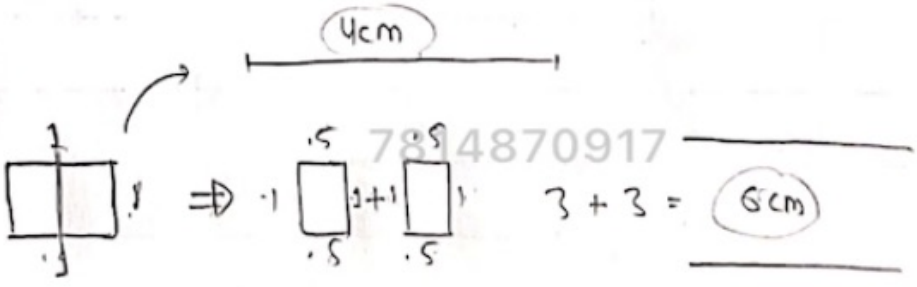
- Particle size can be determined by — Microscopy, sieving method, Light scattering method etc....

→ **Particle shape** - The geometry or (appearance) of a particle when observed under a microscope.

eg. Spherical (round), Cubical (cube shape).

- it affects the (flow) properties.
- Sharp-edged / flat particles may dissolve slower.
- Spherical particles pack better in containers.

Surface Area



CARNIVEL PHARMA
7894870917

Surface Area - The total area exposed on the outer surface of a particle.

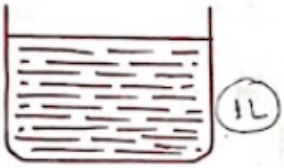
- Smaller particle size → larger surface area per gram.
- Larger surface area → better dissolution and absorption.
- BET Method (Brunauer - Emmett - Teller) - most common method

Flow Properties - It is describe how easily a powder or granules can move or flow, especially during Tablet compression, Capsule filling.

- depends upon particle size, shape and density, hygroscopicity (absorbed moisture)
- affects formulation and efficacy.
- Flow properties ↑ → Production ↑, Mixing ↑
- Flow rate ↓ → Uniformity ↓
- Flow properties determined by Angle of Repose, Carr's index, Hausner ratio etc...

Hygroscopicity - It is the ability of a substance to absorb moisture from the air. Some drugs absorb water quickly and become sticky or degrade.

Density - It is defined as the mass per unit volume of a material. It is especially important for powders and granules, affecting formulation, mixing, compaction and filling processes.



$$\text{Density} = \frac{\text{Mass (m)}}{\text{Volume (V)}}$$

density

$$\rho = \frac{M}{V}$$

M — mass

V — volume

Bulk density (poured density) - Mass of powder divided by bulk volume (including interparticle spaces).

- Used to assess packability and flow properties of powders.

$$\text{Bulk Density} = \frac{\text{Mass}}{\text{Bulk Volume}}$$

$$B.D = \frac{M}{B.V}$$

True Density (Tapped density) - Mass of particles divided by the volume excluding pores.

- Measured using a helium pycnometer.
- Important in tablet formulation and dissolution.

True volume

$$\text{True Density} = \frac{\text{Mass of solid}}{\text{True volume (no voids)}}$$

$$T.D. = \frac{M}{T.V}$$

Angle of Repose - It is the maximum angle formed between the surface of a pile of powder and the horizontal plane, when the powder is allowed to flow freely to form a heap.

- It is an indirect measure of the flowability of powders.

$$\tan\theta = \frac{h}{r}$$

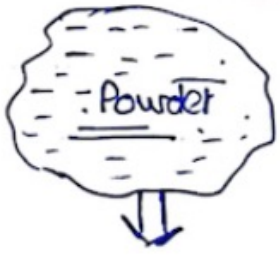
where, h = height of pile.
r = Radius of pile.
θ = Angle of repose.

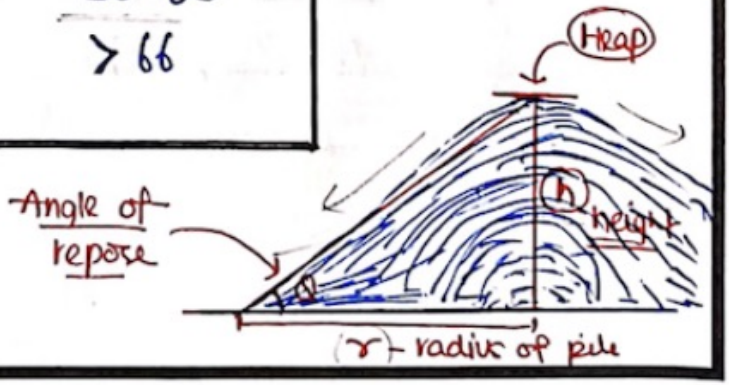
or $\theta = \tan^{-1}\frac{h}{r}$

Angle of Repose ↓ = Flow property ↑.

- It helps us to check the flow nature of powders.

Flow Properties	Angle of Repose
Excellent	or < 25-30
Good	31-35
fair (aid not needed)	36-40
Passable - may hang up	41-45
Poor - must agitate, vibrate	46-55
Very poor	56-65
Very, very poor	> 66





CRISWELL PHARMA

Carr's Index → measure of (compressibility) - It is an indication of powder flowability based on the difference between tapped and bulk densities.

$$\text{Carr's Index (CI)} = \frac{\text{Tapped Density (T.D.)} - \text{Bulk Density (B.D.)}}{\text{Tapped Density (T.D.)}} \times 100$$

Hausner Ratio - It is the ratio of Tapped Density to Bulk Density, and it reflects the inter-particle friction.

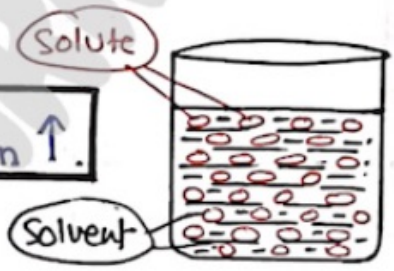
$$\text{Hausner Ratio} = \frac{\text{Tapped Density}_{T.D.}}{\text{Bulk Density}_{B.D.}}$$

Flow property	Carr's Index	Hausner's Ratio.
Excellent	≤10	1.00 - 1.11
Good	≤ 11-15	1.12 - 1.18
Fair	16-20	1.19 - 1.25
Passable	21-25	1.26 - 1.34
Poor	26-31	1.35 - 1.45
Very poor	32-37	1.46 - 1.59
Very very Poor.	>38	>1.60

Careweel Pharma
7814870917

Solubility Profile - It is the concentration of a solute that can be dissolved in a solvent at a given temperature and pressure to form a homogeneous solution.

- Determines bioavailability (poorly soluble drugs – poor absorption).
- Affects stability and shelf life.



Solubility ↑ – Absorption ↑.

(a) **Ionization constant (pKa)** - The ionization constant (pKa) is the pH at which 50% of the drug molecules are ionized and 50% are unionized.

- It is derived from the acid dissociation constant (Ka).

$$pKa = -\log_{10} Ka$$

Eg. $CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

(Unionized) (Ionized form)

- Ionized form :- more water soluble, less lipid soluble.

$H_2O \rightleftharpoons H^+ + OH^-$ → Hydrophilic → Absorption ↓

CAREER PHARMA 7814870917

Unionized form - more lipid soluble, better membrane permeability
→ Lipophilic
→ Absorption ↑

- pKa can be calculated using the Henderson - Hasselbatch equation.
- For acidic drugs :-

$$pH = pKa + \log \frac{\text{Ionised}}{\text{unionised}}.$$

- For Basic drugs :-

$$pH = pKa + \log \frac{\text{Unionised}}{\text{Ionised}}.$$

Example	Ionization	pKa.
very weak acid	unionized at all pH	> 8.
Moderately weak acid	unionized at gastric pH 1.2	2.5 - 7.3
Strong acid	Ionized at all pH	< 2.5.
very weak base	unionized at all pH	< 5.
Moderately weak base	unionized at intestinal pH	5 - 11
Strong base	Ionize at all pH	> 11.

CAREWELL PHARMA

7814870917

(b) pH - Power of Hydrogen OR Potential of Hydrogen.

- pH is the negative logarithum of the hydrogen ion concentration in a solution.
- used to check that particles are acidic / Basic.
- Influences solubility, stability, ionization and absorption of drugs.

$$pH = -\log [H^+]$$

Eg. Solubility of aspirin can be enhanced by addition of alkaline buffer.

- **Dissolution** :- It is the process by which a solid drug dissolves in a solvent to form a solution.
- It is a rate process (unlike solubility which is an equilibrium property).

Solubility ↑ = Solution ↑ = Absorption ↑

- Directly linked to drug absorption and bioavailability.
- Dissolution testing = quality control tool for tablets & capsules.

CARENEL PHARMA

- **Common Ion Effect** - The suppression of the solubility of a weak electrolyte by the presence of a common ion from a strong electrolyte.
- Reduces dissociation of drugs, preventing precipitation.
- Used to control drug release by adjusting solubility.

↓ Decrease solubility of particles.

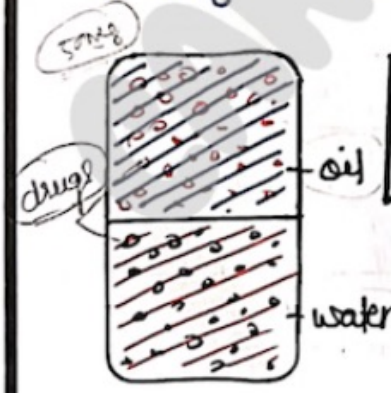
eg. Solubility of weak acid salt (eg. sodium benzoate) decreases in presence of HCl (common ion H^+).

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$ $\quad$ $CH_3COONa \rightleftharpoons CH_3COO^- + Na^+$

common ion

(c) **Partition coefficient** - The ratio of concentration of a drug in oil phase (lipid) to its concentration in aqueous phase (water) when the two are in equilibrium.

- Usually measured in octanol - water system.



$$K_{o/w} = \frac{X_o}{X_w}$$

where, X_o = (conc.) of drug in organic (oily phase)

X_w = (conc.) of drug in aqueous phase.

$K_{o/w}$ > Lipophilic (1)

$K_{o/w}$ < 1 → Hydrop

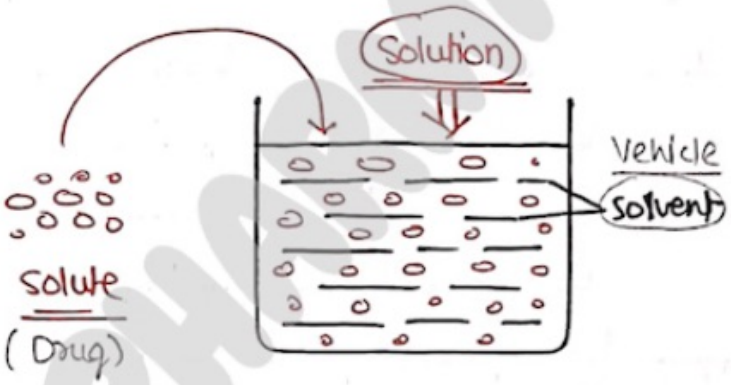
- Drug molecules having higher ($K_{o/w}$) will cross the lipid cell membrane.
- It is used to correlate drug structure with biological activity.

Q. Method of Enhancing Solubility of drugs.

Solubility → It is the ability of (solute) to dissolve in a solvent to form solution at constant temp°.

- there are several (techniques), which are used to increase solubility :-

i) pH change
ii) Co-solvent
iii) Particle-size reduction
iv) Solid dispersion
v) Hydrotrophy method
vi) Complexation



① **pH Change** → A solubility of poorly soluble (drug) that is either a weak base or weak acid may altered by adjusting the (pH) of the solution.

- pH adjustment for improving the solubility in two ways :-
  - Salt formation
  - Addition of (buffer) to the formulation.

② **Co-solvent** → It is a technique to increase the solubility of poorly soluble (drug) in a Liquid. By using co-solvent, we can increase the solubility of poorly soluble drugs. (eg). 2-Pyrrolidine-Propylene glycol.

③ **Particle-size reduction** → The solubility of a product is (S.A↑) also depend on a particle size,

- So, by decreasing the particle size we can increase the solubility of product.

  Particle size ↓ ⇒ Surface area ↑ ⇒ Solubility ↑

- Particle size can be reduced by using microionisation, nano-suspension, grinding etc..

(iv) Solid dispersion → In solid dispersion, A poorly soluble drug dispersed in a highly soluble solid hydrophilic matrix which enhance the solubility.

eg. Hydrophilic carrier.

(v) Hydrotropy → In this method, By adding large amount of secondary solute increase the aqueous solubility of water insoluble drugs.

eg. Solubilization of theophylline (drug) with sodium benzoate (additives) or sodium acetate.

(vi) Chelation → It is a process of association of two or more molecules to form a non-bonded entity. It is used to improve bio-availability of poorly soluble drugs.

eg. Chelates – EDTA etc..

CARESS PHARMA

7814870917

Polymorphism - It is the ability of a solid material (drug or excipient) to exist in two or more crystalline forms with different arrangements of molecules in the crystal lattice but having the same chemical composition. polymorph

α-form → stable, less soluble; β-form → more soluble

- Compounds which show polymorphism are chloramphenicol, indomethacin, sulphonamide, barbiturates show different polymorphic forms in their solubilities, stability and also their pharmacological activities.

  Eg. Carbon exists as diamond and graphite (allotropes).

Ritonavir (used in HIV treatment) ⇒ form I → stable & soluble, form II → less soluble, more stable

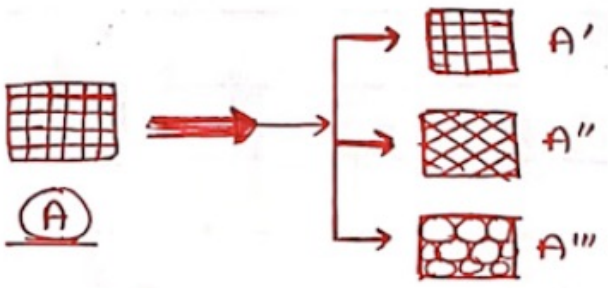
Types :- They are of two types :-

(i) **Enantiotropic polymorphism** - one polymorph can reversibly transform into another depending on temperature or pressure.

Eg. Sulfur

(ii) **Monotropic polymorphism** - one polymorph is stable, others are metastable and irreversibly convert into the stable form.

Eg. Glyceryl stearates.



Three different form

Anyone can be more stable

Chemical Properties

These studied are to know how stable a drug is and how it reacts with environment or excipients.

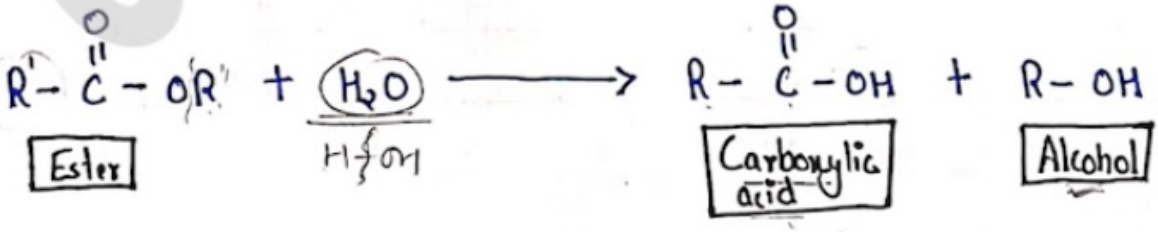
- This help to design a safe and stable dosage form.

(i) Hydrolysis
(ii) Oxidation
(iii) Reduction
(iv) Racemization
(v) Polymerization.

(i) **Hydrolysis** - Most common type of drug degradation.
The decomposition of drug molecules due to reaction with water and get hydrolysed (breakdown of drug substances).

7814870917

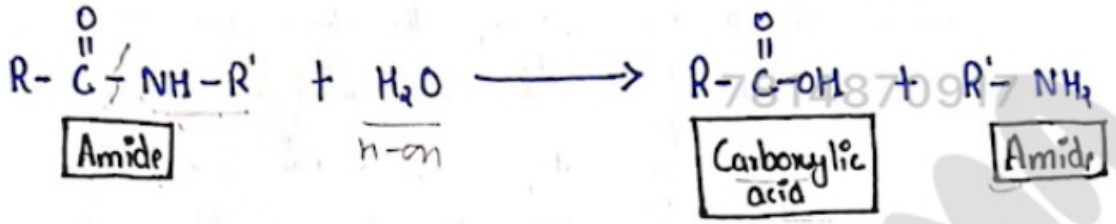
- Mostly drugs with function group ester, amide, lactam, lactone undergoes hydrolysis.
- **Hydrolysis of Ester** -



eg. drugs which undergoes hydrolysis -
Aspirin, Atropin, procaine etc...

CARMEL PHARMA

- **Hydrolysis of Amide** -



Eg. Barbiturates, Dibucaine etc...

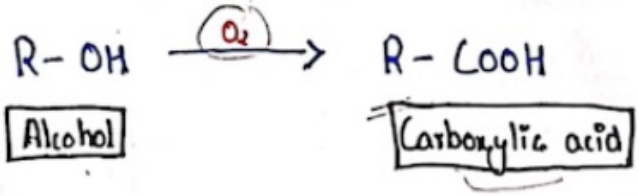
Prevention -

- Use of buffers to maintain optimum pH.
- Store in dry conditions (desiccants).
- Use non-aqueous solvents or prepare dry powders for reconstitutions.
- Cold storage (refrigeration).

(ii) **Oxidation** - Degradation by reaction with oxygen (loss of electrons or addition of oxygen). Often involves free radical chain reactions.

- Oxidation occurs due to oxygen, Heavy metals ions, Light.
- Aldehyde, alcohols, phenols, alkaloids and fats and all are more susceptile to oxidation.

Eg. Rancidity oil.



7814870917

Prevention -

- Use of antioxidants: ascorbic acid, sodium metabisulfite, tocopherol.
- Chelating agents [EDTA] to remove trace metals.
- Replace air with inert gases (nitrogen, CO_2).
- Use amber or airtight containers.
- Store in cool environment.

(iii) **Reduction** - It is a chemical reaction in which a molecule gains electrons or loses oxygen or gain hydrogen.

- It is the opposite of oxidation.

eg. Chloramphenicol → gets reduced at nitro group to form inactive compounds.

Prevention -

- Store under controlled pH and temperature.
- Use antioxidants / reducing inhibitors if suitable.
- Avoid combination with strong reducing excipients (like sugar, polyols) unless safe.

CAREWELL PHARMA

(iv) **Racemization** – It is an conversion of an optically active drug (enantiomer) into its mirror image (opposite enantiomer). (R/S)

- This may reduce or completely change drug activity.
- alter pharmacobinetic, pharmacological and toxological properties.
- It depends on temperature, solvent, catalyst and presence or absence of light.

eg. L-Adrenaline (active) is 15-20 time more active than D-Epinephrine.

Prevention –

- Store under controlled pH and temperature.
- Use stabilizers if necessary.

(v) **Polymerization** – It is a chemical reaction where two or more drug molecules combine to form high molecular weight products.

- Often leads to loss of potency and sometimes toxicity.

eg. PABA (para-aminobenzoic acid) derivatives.

Prevention :-

- Proper choice of solvent and stabilizers.
- Use of controlled storage conditions.

CARSWELL PHARMA

Application of Preformulation Consideration in Dosage form Development

1. Solid Dosage Form (Tablets, Capsules, Powders).

Preformulation studies help to select proper excipients and ensure stability, bioavailability and manufacturability.

- **Solubility & Dissolution -**
  - Drug with poor solubility → need solubility enhancement (eg. salt formation).
  - Ensures proper absorption.
- **Particle Size & Surface Area.**
  - Small particle size → higher dissolution, better absorption.
  - Important for uniform blending of powders.
- **Polymorphism & Crystallinity.**
  - Different forms may have different solubility and stability.
  - Example - Ritonavir tablet failed due to unstable polymorph. form-II
- **Flow Properties (Angle of Repose, Compressibility).**
  - Important for tablet / capsule manufacturing.
  - Prevents problems like weight variation.
- **Stability Studies.**
  - Protects from hydrolysis, oxidation, light.
  - Example - Aspirin needs dry packaging due to hydrolysis.

PHARMA

2. Liquid Dosage form (Solution, Suspension, Syrups, Emulsions)

Here drug solubility and stability are most important.

- **Solubility & pH Adjustment.**
  - Drug should be soluble in chosen solvent (water, alcohol, glycerin).
  - pH adjustment buffers help to keep drug stable.
- **Partition Coefficient & vehicle Selection.**
  - Helps in deciding co-solvents (ethanol, PEG, propylene glycol).
- **Stability Against Hydrolysis / Oxidation.**
  - Antioxidants (ascorbic acid, sodium bisulfite) used.
  - Light-sensitive drugs packed in amber bottles.
- **Viscosity Consideration.**
  - For suspensions / syrups, viscosity agents (CMC, PVP) added for uniformity
- **Preservatives.**
  - To avoid microbial growth in aqueous liquids (eg. parabens, benzoates)

3. Parenteral Dosage Forms (Injections, IV Infusions)

Preformulation is critical because these are directly given into the body → require highest purity, sterility and stability.

- **Solubility & Isotonicity.**
  - Drugs must be soluble and isotonic with blood (0.9% w/v NaCl).
  - Adjusted with sodium chloride, dextrose.

7814870917

- **pH and Buffering.**
  - Optimum (pH) for solubility + Stability + compatibility with body tissues
  - Example - pH (3-9) usually acceptable.
- **Stability in solution.**
  - Must be stable during (storage) and (sterilization).
  - Lyophilization (freeze-drying) used for unstable drugs.
- **Sterility & Pyrogen Testing.**
  - Preformulation ensures drug is compatible with sterilization methods (autoclaving, filtration).
- **Container Compatibility.**
  - No leaching / interaction with glass or plastic vials.
  - Example - Adsorption of insulin on plastic containers avoided by adding (albumin).

CAREWELL PHARMA
7814870917

Q-2

BCS - Biopharmaceutical Classification System

- The Biopharmaceutical classification system (BCS) is a scientific framework introduced by Amidon et al. (1995).
- It is used to differentiate the Drug on the basis of their solubility and permeability.
- It helps in understanding drug dissolution, absorption and bioavailability which are critical for oral dosage forms
- It's main purpose is to predict oral absorption, guide formulation design and help in bioequivalence waivers (biowaivers) by regulatory agencies like US FDA (eg: Class-I only used in-vitro)
- It classifies drug substances into four classes based on their aqueous solubility and intestinal permeability.

	Solubility	Permeability	Example
Class I	High	High	Metoprolol, Paracetamol
Class II	Low	High	Phenytoin, ketoprofen.
Class III	High	Low	Ranitidine, Cimetidine.
Class IV	Low	Low	Furosemide, Hydrochlorothiazide

Carewell Pharma

(i) Class I - These compounds are well absorbed and their absorption rate is usually higher than excretion

- make tablets or oral dosage form easily.

Eg. Paracetamol. (High solubility, High permeability)
Metoprolol etc

(ii) Class II - These compound have good absorption but slow dissolution and low solubility.

Eg. Phenytoin. (Low solubility, High permeability)
ketoconazole etc.

(iii) Class III - These compounds have good solubility but low absorption due to low permeability.

Eg. Cimetidine, (High solubility, Low permeability)
Atenolol, Ranitidine etc.

(iv) Class IV - These compounds have low absorption, low solubility

- Poor bioavailability:-

Eg. Furasemide, (Low solubility, Low permeability).
Hydrochlorothiazide etc..

Criteria for BCS classification

(a) **Solubility** - A drug is highly soluble if the highest single dose can dissolve in ≤ 250 ml aqueous medium across a pH range of 1.2 - 7.5.,

- Low solubility drugs → need solubility enhancement techniques.

CAMPUS PHARMA

(b) **Permeability** - A drug is considered highly permeable if ≥ 90% of the administered dose is absorbed (based on mass - balance or compared with IV dose).

(c) **Dissolution** - A drug product is considered rapidly dissolving if ≥ 85% of the drug dissolves within 30 minutes in 900 ml buffer at pH 1.2, 4.5 and 6.8.

Significance -

- Saves time and cost in drug approval.
- Ensures consistent drug release and absorption in body.
- Helps in deciding formulation approaches.
- Stimulate food effects on bio-availability.
- Used to develop quality control.
- Used to study **IVIVC**
  - → In Vitro In Vivo correlations.
    - In Vitro → outside the body → check dissolution/solubility
    - In Vivo → inside the body. → Absorption + Bioavailability

Stability Analysis

Stability is a stage / state / quality being steady and not changed.

- But there are many factors which can affects the stability of

of drug substances.

- Some physical, chemical and microbiological changes influence the effectiveness, safety and stability of final drug products.

Physical change - Change in appearance of drug products and also their M.P, B.P, purity, clarity and other physical factors.

Chemical change - degradation of drug substances.
Eg. oxidation, hydrolysis etc...

Microbial changes - drug substance can be decompose due to micro-organism (microbial contamination).

- The physical, chemical, microbiological, therapeutical, toxicological properties of any drug is remain constant form as per begining → **Stability**
- These studied are conducted to recommend storage conditions, establish shelf-life of drugs.

⇒ **Stability analysis includes**

(a) Solid-state stability analysis.
(b) Solution-state stability analysis.
(c) Compatibility studies.

7814870917

(a) **Solid - state stability Analysis** - In this, we study about the physical or chemical properties of the drug molecules that may affects the stability of a drug products.

- In this we study about solid dosage form and those factors which affects solid dosage form such as temperature, pH, humidity, hydrolysis, oxidation etc...

(b) **Solution - State stability Analysis** - Degradation in solution form is more rapid than solid form.

- In this, we study about the stability of drug substance at various factors which affects stability such as pH, temp etc..
- The effect of pH on stability is important in the development of both oral and parenteral dosage form.

7814870917

(c) **Compatibility studies** - In this we study about the drug - excipients interaction and how they affects the stability of drug substances.

CAREWELL PHARMA